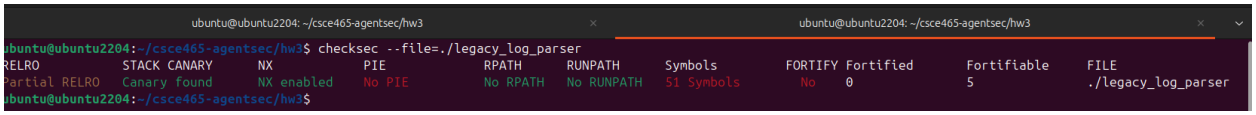
Solo (I thought I would benefit more from doing it alone, but I just ended up not being able to complete)
Caleb Bui
134001202

Task 1:



Checksec for task 1, although I won't need until Task 3

```
(gdb) info frame
Stack level 0, frame at 0x7fffffffdc00:
 rip = 0x4013ef in vulnerable_copy (legacy_log_parser.c:57); saved rip = 0x401557
 called by frame at 0x7fffffffdc40
 source language c.
 Arglist at 0x7fffffffdbf0, args: input=0x405480 "This is a test log file\n",
    n=24
 Locals at 0x7fffffffdbf0, Previous frame's sp is 0x7fffffffdc00
 Saved registers:
  rbp at 0x7fffffffdbf0, rip at 0x7fffffffdbf8
(gdb) p &buf
$1 = (char (*)[128]) 0x7fffffffdb70
(gdb) info address win
Symbol "win" is a function at address 0x4012b6.
(gdb)
```

Saved return address and address of win,
Offset = start of buf - rip address = 0x88 = 136 bytes

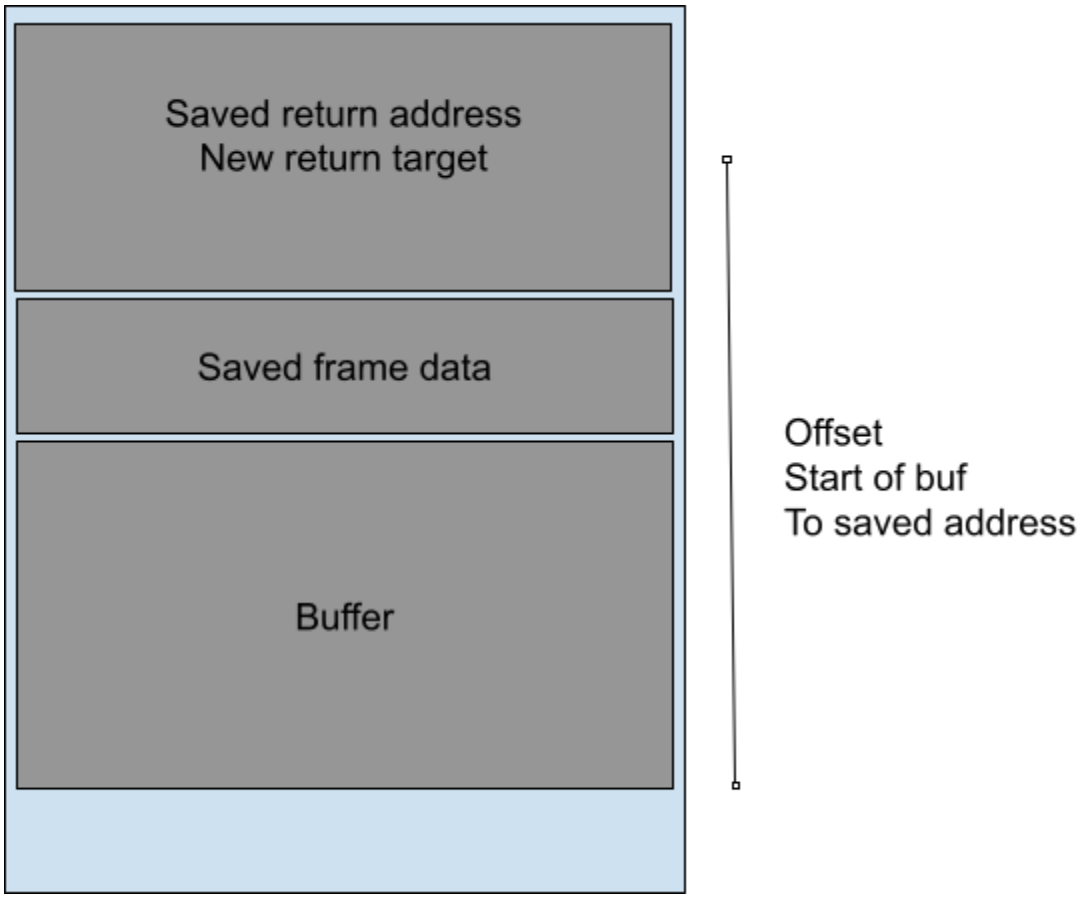
```
[+] Read 136 bytes from test.bin

Breakpoint 1, vulnerable_copy (input=0x405480 'A' <repeats 136 times>, n=136)
    at legacy_log_parser.c:57
57          memcpy(buf, input, n);
(gdb) c
Continuing.
[+] Log contents: AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA

Program received signal SIGBUS, Bus error.
parse_log (
    path=<error reading variable: Cannot access memory at address 0x41414141414141
19>) at legacy_log_parser.c:101
101         free(input);
(gdb) c
Continuing.

Program terminated with signal SIGBUS, Bus error.
The program no longer exists.
(gdb)
```

Failed overflowing with payload to change return address, it overflowed but I don’t think I correctly managed to corrupt the return address and instead just caused a memory error.



Task 2:
No logs, I just didn’t manage my time and ended up not finishing